Working Groups (all)

# GY457 Autumn Term 2025 (Seminar Topic 7)

Lecturer: *Dr Sara Bagagli*

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**Context**[1]

This seminar builds on the lecture on Topic 7: "The vertical dimension of cities." During the lecture, we have analysed vertical rent gradients within buildings theoretically and empirically. We have also investigated how per-unit construction costs that increase in height as well as planning regulations determine the urban height profile. The purpose of this seminar is to deepen the understanding of these concepts and to explore how the empirical evidence can help identifying the profit maximizing building height for a tall building.

**"Homework" to be carried out prior to the seminar**

The compulsory readings for this seminar are:

**Ahlfeldt, G. M., and McMillen, D. (2018): Tall Buildings and Land Values: Height and Construction Cost Elasticities in Chicago, 1870 – 2010.** The Review of Economics and Statistics. 100(5), 861.875.

**Cheshire, P. C., and Dericks, G. (2020) 'Trophy Architects' and Design as Rent-seeking: Quantifying Deadweight Losses in a Tightly Regulated Office Market**. Economica. 87. 10781104

**Liu, C., H., Rosenthal, S. S., and Strange, W. C. (2018): The vertical city: Rent gradients, spatial structure, and agglomeration economies.** Journal of Urban Economics. 106. 101-122.

All students are expected to read these papers. All students must be prepared to answer the questions during the seminar. It is recommended that they execute the numerical exercise and prepare some notes. However, no submission of written answers is expected. *Presenters*, in addition, are expected to prepare slides for a joint presentation of their answers (one presentation per seminar group). In doing so, they may split the tasks or work together on all tasks as they prefer. Presenters are advised to have a version of the Excel file "GY457_Topic_7_Seminar_Worksheet.xlsx" ready on the seminar computer so that the effects of varying parameter values can be evaluated if necessary.

*Seminar tasks:*

Imagine that you (the presenters) represent a developer considering buying a vacant plot of land, with the aim of developing a tall office building and rent out the space over a very long period. In the area, the planning system has imposed a 10-floor height limit, but you have been told from a credible source that the planner will be willing to negotiate the height limit if you hire a "Trophy architect". You are bidding against other developers for the plot of land.

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[1] The seminar was originally designed by Prof. Ahlfeldt and is kindly used with permission.

Your aim is to be successful in the bidding under the condition that the investment gives you a rate of return of 5%. In doing so, you must decide on how tall to build and whether to hire a "Trophy architect".

1) Discuss how, theoretically, a developer can find the profit maximizing building height in the absence of planning regulation. What are the main factors pushing for a higher and lower optimal building height? How do these parameters and the resulting optimal building height depend on land use?
2) The worksheet "Parameters" in the file "GY457_Topic_8_Seminar_Worksheet.xlsx" provides you with additional parameters describing the scenario. Some parameters are missing but can be borrowed from the literature. Discuss your choice of the appropriate parameters in some detail. Given the parameters and your choices, what is the profit-maximizing building height in the absence of height restrictions in the given scenario? What is the net present value of the expected rental revenue? What is the expected construction cost? Which price would you be willing to bid for the land parcel?
3) Now assume the 10-floor height restriction is binding (Scenario 1). What is the net present value of the expected rental revenue? What is the expected construction cost? Which price would you be willing to bid for the land parcel?
4) Now assume that the planner is willing to allow for extra height in return for you hiring a trophy architect. With reference to evidence, discuss how a trophy architect might be characterized, what percentage cost the trophy architect would add to the total cost of a building, and how many additional floors the planner would likely be willing to grant planning permission for. How much would you be willing to bid in this scenario, and would you hire the trophy architect?
5) Scenario 2: What if instead of a 10-floor height limit there was a 40-floor height limit? What land price would you be willing to bid, and would you hire a trophy architect?
6) How would the optimal building height change if
    a. The height elasticity of rent increased?
    b. The height elasticity of cost increased?
    c. The baseline construction cost and the baseline rent increased both by 25%?
    d. If the baseline rent was expected to increase at a rate of 2% per year? Would you hire a trophy architect in the scenario with a 10-floor height limit and a 40-floor height limit?

**During the seminar**

After an introduction and discussion of initial questions, the presenters will walk the group through their answers to the questions in an "interactive" presentation. All analytical and numerical calculations should be laid out in a transparent and accessible manners. All presenters should carry an approximately similar weight in the presentation. During the presentation, it is expected that the presenters "engage" with the audience, i.e. prompt the audience to suggest answers. Likewise, the audience is expected to challenge the answers provided by the presenters.

The presentation, in total, may cover up to 60 minutes. The lecturer will monitor the presentation and the discussion and intervene if appropriate. The presenters are expected to make notes during the discussion in case corrections to their suggested answers are due.

**After the seminar**

If indicated by the feedback during the seminar, presenters will revise their presentation. Within 24 hours, they will then ask the instructor to upload the slides onto the GY457 Moodle page. Please, use the following filenames to save your presentations: Topic_7_Group_X, where X indicates the seminar group.

**Guide to numerical solution**

Developers' profit maximizing problem leads to the following optimal height (adapted from Ahlfeldt and Barr, 2022; Ahlfeldt and McMillen, 2018):[2]

$$S^* = \left(\frac{p_u(1+\omega)}{c_u(1+\theta)}\right)^{\frac{1}{\theta-\omega}}$$

Where:

- $p_u$ is expressed as the present value of rent of 1-floor building, i.e. Net Operating Income/Internal Rate of Return (NOI/IRR)
- $c_u$ is the unit cost of 1-floor building
- $\theta$ is the elasticity of average per square meter floor space construction cost wrt height
- $\omega$ is the elasticity of average per square meter floor space rent wrt height

Once the optimal building height is computed, we can get profits as:

$$\pi^* = p_u {S^*}^{1+\omega} - c_u {S^*}^{1+\theta} - \frac{r}{l}$$

Where:

- r is land rent
- l is ground floor area

To make an offer, the profits must be non-negative. This condition should give the maximum bid a developer is willing to make for that parcel of land.

[2] This equation can be derived from the developers' profit maximizing problem: $\pi = pS - cS - \frac{r}{l}$, and considering $p = p_u S^{\omega}$, $c = c_u S^{\theta}$. Where p, c are baseline values for a one-story building, and ω, θ are respectively the elasticities of floor space rent and construction cost with respect to height.